

An Empirical Study of the Conformal Information Gap

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Abstract

Every forecasting architecture restricts the information available to its predictive law. Under logarithmic loss the irreducible cost of replacing the full input by a retained representation is exactly the conditional mutual information between outcome and input given the representation. Following a companion note, we call it the information gap. For a single pooled residual law the gap is the mutual information between residual and input, and pooling after any invertible state-dependent change of coordinates has the analogous gap in the transformed coordinates. Conformal calibration can improve estimation and provide a coverage certificate, but it cannot recover predictive information excluded by the retained representation. We apply the decomposition to a nested grammar of online transformations on 701 economic series. Conditional scale is the largest and most reliable component of the measured architecture gain, and, holding the point predictor fixed, scale-normalized empirical pooling beats raw pooling on 87 percent of the series under a quantile approximation to CRPS. The value of residual pooling depends jointly on the coordinates, the residual-law estimator, and the scoring rule.

1 Introduction

This paper asks when residual pooling is an effective component of probabilistic forecasting. The motivating application is an online distributional time-series forecaster built from chained transformations, but the underlying question is general: what predictive value is lost when a forecast uses only a representation $T(X)$ of the available information X ? Under logarithmic loss that loss has an exact decomposition, and we call its irreducible representation component the *information gap*. A recent note, blinded for review, introduces the gap for a single pooled residual law. This paper extends the definition to arbitrary retained representations and keeps the treatment self-contained for convenience.

A forecasting method is partly a model and partly a decision about what information the forecast is allowed to use. A global residual distribution remembers no input-dependent residual state. A volatility model remembers an estimated scale. A tree ensemble remembers a partition and its fitted leaf values. A state-space filter remembers a posterior state. A neural forecaster remembers a learned embedding. In each case the predictive law is measurable with respect to a representation $T(X)$ of the available information X .

We ask a simple question:

What is the predictive cost of replacing the full information X by a representation $T(X)$?

Under logarithmic score the answer is exact. Let Z be the target, let $p(z|x)$ denote the true conditional density, and let $q(z|T)$ be any predictive density that can depend on X only through T . Then

$$\mathbb{E} \log \frac{p(Z|X)}{q(Z|T)} = \underbrace{\mathbb{I}(Z; X|T)}_{\text{representation regret}} + \underbrace{\mathbb{E}_T \text{KL}(P_{Z|T} \| Q_{Z|T})}_{\text{within-representation regret}}. \quad (1)$$

Here $\mathbb{I}(Z; X|T)$ is the conditional mutual information between Z and X given T , $P_{Z|T}$ is the true conditional law of Z given T , and $Q_{Z|T}$ is the forecast. Formal definitions are collected in Section 4. The first term is the Bayes value of the information in X that is absent from T . We call it the *information gap*, or *floor*, of the representation. It vanishes precisely when T is sufficient for predicting Z . The second term measures how far the fitted forecast lies above the best forecast permitted by the architecture. Global residual pooling deliberately uses a coarse residual representation. The question is whether its reduced estimation burden compensates for the resulting representation floor.

Identity (1) is classical in substance. It is a conditional relative-entropy decomposition, closely related to value-of-information results, Blackwell sufficiency, and proper-scoring-rule entropy (Lindley, 1956; DeGroot, 1962; Blackwell, 1953; Grünwald and Dawid, 2004). It is also the misspecified-model decomposition of White (1982). The model class is the family of all predictive laws measurable with respect to T , and its pseudo-true member is $p(\cdot|T)$. For such an information-restricted class the irreducible misspecification error takes the closed form $\mathbb{I}(Z; X|T)$.

To our knowledge this decomposition has not been stated for conformal prediction. It applies there with unusual ease, because residual pooling fixes T by construction. The irreducible term becomes the mutual information $\mathbb{I}(R; X)$ between residual and input. We state the decomposition for pooled residual systems and, in Theorem 11, for pooling in any state-dependent coordinates. We then price fitted architecture increments against the decomposition on a large benchmark, and we separate the part of a residual method’s predictive shortfall due to its coordinates from the part due to its estimator. The accounting applies to any forecasting architecture. Conformal prediction is the running case study because its representation choice is explicit.

We now specialize. Let the target be a real-valued outcome Y , fix a point predictor $\hat{\mu}(x)$, and define the residual

$$R = Y - \hat{\mu}(X).$$

A single-shape residual forecast uses one law H for R at every input and reports

$$q_H(y|x) = h(y - \hat{\mu}(x)).$$

The point forecast may be highly adaptive, but the residual-shape representation is trivial. Conditional on the fixed location model, the best input-blind residual law is the marginal law P_R , and its log-score regret relative to the conditional residual oracle is

$$\mathbb{I}(R; X).$$

Signed-residual conformal predictive systems (CPS) converge to this marginal residual law under standard conditions (Vovk et al., 2019). Thus they are population-optimal within the input-blind residual class. The representation floor remains. Ordinary absolute-residual intervals impose an additional symmetrization restriction.

The distinction between representation and calibration is central. A conformal procedure can change a fitted CDF, quantiles, intervals, and therefore proper scores. It may improve or

worsen the estimation term. What it cannot do is reduce $I(R; X | T)$ while leaving T unchanged. A conformal method reduces this floor only when its score, normalization, partition, localization rule, or learned embedding actually enlarges the retained representation.

The same decomposition gives a precise architecture-selection criterion. If T_A is a coarsening of T_B , the oracle advantage of T_B is the additional predictive information $I(Z; T_B | T_A)$. At finite samples this gain competes with the richer method’s extra within-representation excess risk. Under a prior on Nature, the richer method wins exactly when its information advantage exceeds its statistical cost. This yields a concrete theoretical explanation for why boosted conditional-distribution learners can outperform global residual pooling under priors concentrated on learnable heteroscedastic or interaction structure (Friedman, 2001; Chen and Guestrin, 2016; Duan et al., 2020).

Our claims divide accordingly. The value-of-information lemma, its proper-score analogue, monotonicity under refinement, and the approximation–estimation split are classical or immediate. We state them with explicit regularity conditions and with learned representations treated conditionally on training data. Our own contributions are the transform-pooling theorem, which prices every pooled-residual scheme by the coordinates in which it pools, and two experiments built on it: a nested-ladder ablation over 701 economic series (Figure 1) and a matched-predictor comparison that separates the cost of a method’s coordinates from the cost of its estimator.

2 Related ideas

2.1 Value of information, sufficiency, misspecification, and proper scores

Under log loss, the reduction in Bayes risk obtained by observing additional information is a mutual information. This is classical (Lindley, 1956; DeGroot, 1962). Blackwell’s comparison of experiments formalizes when one information structure is more useful than another for every decision problem (Blackwell, 1953). For logarithmic prediction, the value of a refinement has the explicit form of conditional mutual information.

The same identity has ancestors in at least three further literatures. In misspecified-model theory, maximum likelihood converges to the KL projection of the truth onto the model class (White, 1982). Here the class is the set of T -measurable predictive laws. The pseudo-true member is $P_{Z|T}$, and the projection distance is $I(Z; X | T)$. In sufficient dimension reduction, one seeks $T(X)$ with $Z \perp X | T(X)$ (Li, 1991; Fukumizu et al., 2004). That is exactly the zero-floor condition $I(Z; X | T) = 0$. The decomposition supplies the graded log-score price of insufficiency, in the spirit of information-theoretic reductions that minimize lost mutual information (Globerson and Tishby, 2003). In the information bottleneck (Tishby et al., 1999), the Markov chain $Z - X - T$ gives $I(Z; X) = I(Z; T) + I(Z; X | T)$, so minimizing the floor is maximizing retained predictive information. The bottleneck penalizes representation complexity by $I(X; T)$. We penalize finite-sample estimation and certification cost instead, and the two need not coincide.

Akaike’s programme of choosing the fitted model with smallest estimated expected KL discrepancy (Akaike, 1974) likewise separates approximation error from estimation optimism. Our refinement is that for information-restricted classes the approximation term is identified exactly.

We apply these facts to the design of forecasters. Here T is the state a forecasting method retains, not a statistic an analyst chooses afterward. The decomposition then separates two errors: the error built into that choice of state, and the error of fitting within it.

2.2 Calibration and sharpness

The forecasting literature distinguishes calibration from sharpness and advocates maximizing sharpness subject to calibration (Gneiting et al., 2007; Dawid, 1984). Proper scores combine these considerations into a single decision criterion (Gneiting and Raftery, 2007). Conformal predictive systems provide calibrated predictive distributions (Vovk et al., 2019), while split conformal methods provide set-valued coverage guarantees (Vovk et al., 2005; Lei et al., 2018; Angelopoulos and Bates, 2023). The present decomposition identifies an additional quantity: even the best calibrated or best-fitted forecast measurable with respect to a fixed T cannot improve upon the representation floor.

2.3 Adaptive conformal prediction

Mondrian methods, normalized scores, localized conformal procedures, conformalized quantile regression, learned conformity scores, and temporal conformal methods all permit the report to depend on more input-derived information than global residual pooling (Boström et al., 2021; Lei et al., 2018; Romano et al., 2019; Chernozhukov et al., 2021; Gibbs and Candès, 2021; Auer et al., 2023). In the present language they choose different representations, though they are not generally arranged in a total order. When one representation refines another, the reduction in oracle log-score regret is exactly the predictive information captured by the refinement.

2.4 Conditional-coverage impossibility

Distribution-free exact conditional coverage is impossible in rich nonatomic settings without trivial or infinitely wide sets (Lei and Wasserman, 2014; Foygel Barber et al., 2021). Approximate and subgroup-conditional guarantees interpolate between marginal and pointwise conditional validity (Vovk, 2012; Gibbs et al., 2025; Braun et al., 2025). These results concern the statistical feasibility of certification at fine conditioning resolution. They complement, but are not identical to, the information decomposition. A common scalar “frontier” requires an additional complexity measure for the representation, such as cell mass, VC dimension, effective local sample size, or description length.

3 Measuring representation value: an empirical ladder

This section measures the practical value of progressively richer retained states. It first compares complete forecasting architectures, and it then holds the point predictor fixed to isolate the cost of residual-pooling coordinates. The floors $I(Y; X | T)$ are population quantities, so we measure realized gains instead. Identity (1) supplies the measurement device. We present the measurements before the supporting theory, and the identities they rest on are one-line consequences of (1), stated with proofs in Sections 4–7, with forward references marked.

The design borrows its ingredients from forecasting practice, which has evolved a grammar of transformations (location adjustment, scale normalization, differencing, seasonal adjustment, autoregressive state, nonlinear coordinates), each retained because it improved earlier forecasting benchmarks. We arrange this grammar as a ladder of forecasting architectures indexed by a rung j , running from $j = 0$ for the trivial architecture to $j = m$ for the full grammar, in which each rung up to $j = m - 1$ differs from its predecessor only by admitting more transform families.

The final rung is the complete implemented forecaster, and it changes more than the candidate pool, as discussed below.

Each rung works as follows. A candidate forecaster is a chain of online transformations applied to the observation: subtract a tracked level, take a difference, divide by a tracked scale, remove a seasonal component, or filter by a fitted autoregression. Each transformation updates its own parameters after every observation, and each is, conditional on its own past state, an invertible change of variables in the current observation. This is exactly the setting of Theorem 11: a predictive law for the whitened quantity at the bottom of the chain pushes back through the inverse transformations, Jacobians included, to a predictive law for the observable. The chain ends in a terminal residual law, a Gaussian or a Gaussian scale mixture with tracked variance, and the rung's forecast is a likelihood-weighted mixture over its pool of such candidates, with weights updated online and geometrically discounted.

We run the rung forecasters side by side on each series. At each time t every forecaster issues its one-step density forecast from its own tracked parameters and ensemble weights, all computed from observations before t , and then it sees y_t and updates. For the theory we take T_j to be everything forecasters 0 through j know just before forecasting, namely their tracked parameters, predictive laws, ensemble weights, and accumulators, so that

$$T_0 \subset T_1 \subset \dots \subset T_m,$$

where each T_j is implicitly the time- t state and t is suppressed except where it matters. The richer representation contains the poorer one by construction, rung j 's forecast $\hat{Q}_j$ uses only T_j , and the decomposition applies. The observed forecasts and scores are unchanged by this accounting. One caution is that the larger pool also re-weights the candidates it shares with the smaller pool, so Δ_j prices the whole of rung j 's extra state and not the new transform family in isolation. For a score S and fitted forecasts $\hat{Q}_j$, define the *measured representation value* of rung j on a benchmark collection as

$$\Delta_j := \hat{\mathcal{L}}(\hat{Q}_{j-1}) - \hat{\mathcal{L}}(\hat{Q}_j), \quad (2)$$

where $\hat{\mathcal{L}}(\cdot)$ is the held-out average of S over the benchmark collection. This is the realized incremental predictive value of adding the j th transform family. When S is the log score we write simply Δ_j , as in the proposition below, and for a general proper score we write Δ_j^S for the same measured difference.

Proposition 1 (Bridge between measured and oracle value). *Condition on the training data D_n and let $\hat{Q}_j$ be a fitted T_j -measurable forecast with within-representation excess risk*

$$\epsilon_j(D_n) = \mathbb{E}[\text{KL}(P_{Y|T_j, D_n} \parallel \hat{Q}_j(\cdot | T_j, D_n)) \mid D_n].$$

Then the population version of (2) satisfies

$$\Delta_j = \text{I}(Y; T_j \mid T_{j-1}, D_n) + \epsilon_{j-1}(D_n) - \epsilon_j(D_n). \quad (3)$$

The proof, a subtraction of Lemma 3 applied at the two rungs, is in Appendix B.

For a general proper score S the excess-risk term changes along with the information term. Define the score-specific within-representation excess and the generalized conditional information

$$\begin{aligned} \epsilon_j^S(D_n) &= \mathbb{E}[d_S(P_{Y|T_j, D_n}, \hat{Q}_j(\cdot | T_j, D_n)) \mid D_n], \\ J_S(Y; T_j \mid T_{j-1}, D_n) &:= \mathbb{E}H_S(P_{Y|T_{j-1}, D_n}) - \mathbb{E}H_S(P_{Y|T_j, D_n}). \end{aligned}$$

Invoking Theorem 8 at the two rungs and subtracting, exactly as in the proof above, the measured difference (2) computed under S satisfies

$$\Delta_j^S = J_S(Y; T_j | T_{j-1}, D_n) + \epsilon_{j-1}^S(D_n) - \epsilon_j^S(D_n).$$

No Shannon-style chain rule is implied for J_S .

Δ_j is the realized incremental predictive value of the rung, information gain net of the change in within-representation excess risk, and nothing stronger.¹

In the online benchmark the conditioning must be chosen with care. Conditioning on the entire preforecast history would fix every rung state and zero the information term. The representation variable is the random past itself, $X_t = \mathcal{F}_{t-1}$ with $T_{j,t} = T_j(\mathcal{F}_{t-1})$. The exact population statement is therefore time-indexed,

$$\Delta_{j,t}^{\text{pop}} = \mathbb{I}(Y_t; T_{j,t} | T_{j-1,t}) + \epsilon_{j-1,t} - \epsilon_{j,t},$$

conditional at most on a fixed initialization. The reported Δ_j is the realized benchmark analogue of the forecast-time average $N^{-1} \sum_t \Delta_{j,t}^{\text{pop}}$. This is the one-step decomposition of Section 9 applied rung by rung.

A finer ledger separates three prices rather than two. Let $\mathcal{Q}_{T,D_n}$ be the model family the method actually fits at representation T , and conditional on D_n let

$$Q_{T,D_n}^\dagger \in \arg \min_{Q \in \mathcal{Q}_{T,D_n}} \mathbb{E}[\mathcal{L}(Q, Y) | D_n]$$

minimize conditional expected risk. Reading every loss below as a conditional expected risk given D_n , and with the dependence of the conditional laws $P_{Y|X}$ and $P_{Y|T}$ on D_n suppressed in the notation,

$$\mathcal{L}(\hat{Q}_T) - \mathcal{L}(P_{Y|X}) = \underbrace{\mathcal{L}(P_{Y|T}) - \mathcal{L}(P_{Y|X})}_{\text{representation cost}} + \underbrace{\mathcal{L}(Q_{T,D_n}^\dagger) - \mathcal{L}(P_{Y|T})}_{\text{model-form cost}} + \underbrace{\mathcal{L}(\hat{Q}_T) - \mathcal{L}(Q_{T,D_n}^\dagger)}_{\text{estimation cost}},$$

and under log loss the first term is $\mathbb{I}(Y; X | T)$. The second and third terms are nonnegative in this conditional expected sense. On a realized test sample an estimated forecast can beat $Q_{T,D_n}^\dagger$ by chance.

The ϵ_j of (3) lump the second and third terms together. The terminal-leaf comparisons of Section 3.2 move the model-form and estimation terms while holding the representation fixed.

Proposition 2 (Cumulative ladder decomposition). *Summing (3) over the ladder, the interior excess-risk terms cancel:*

$$\sum_{j=1}^m \Delta_j = [\mathbb{I}(Y; X | T_0, D_n) - \mathbb{I}(Y; X | T_m, D_n)] + \epsilon_0(D_n) - \epsilon_m(D_n), \quad (4)$$

and the population identity beneath it carries an explicit remainder:

$$\mathbb{I}(Y; X | T_0, D_n) = \sum_{j=1}^m \mathbb{I}(Y; T_j | T_{j-1}, D_n) + \mathbb{I}(Y; X | T_m, D_n). \quad (5)$$

¹In particular, we do not claim that Δ_j is a lower bound on the oracle gain. That would require $\epsilon_j \geq \epsilon_{j-1}$, and there is no generic reason for this excess risk to be monotone in pool size. Adding candidates can increase estimation variance, but it can also reduce error through model averaging, and the two excesses are divergences from different conditional targets. Equation (3) does dictate the correct reading of a small or negative Δ_j : it does not show that the j th family carries no predictive information, only that its information gain did not exceed its within-representation excess here.

The proof, two telescopes and Corollary 6, is in Appendix B.

The remainder term in (5) is not observed: the richest rung is another fitted architecture, not the conditional oracle. The ladder therefore measures the portion of the representation deficit *recovered by the chosen grammar*. The measurement is additive over families and exact up to the two endpoint excess-risk terms. It does not measure the entire floor of the coarsest rung, and a positive measured total does not by itself certify that the population information is positive series by series.

Two conventions are still required. First, the measured quantities are *representation values*, not information quantities: they depend on the finite sample, the estimation procedures, and the benchmark distribution. Second, the split of the total into $\sum_j \Delta_j$ depends on the order in which families are inserted, because conditional gains interact. A decomposition must fix a canonical nesting in advance, or average over insertion orders, and report the convention. With five families, exact Shapley attribution requires only 2^5 subset pools, and it is order-robust though not grouping-robust.

The grammar is not bespoke to this experiment. It is the transform library of Cotton (2026): the established devices of the time-series literature (location, scale, differencing, seasonal adjustment, autoregressive state), retained where they improved prediction on earlier benchmarks. The grammar itself may therefore be read as an empirical approximation to the representation families that perform well under the empirical distribution of forecasting tasks. Its rungs approximate a sequence along which $I(Y; X | T_j)$ decreases. That sequence is an empirical analogue of the information-bottleneck curve (Tishby et al., 1999), with finite-sample estimation cost in place of the bottleneck’s complexity penalty $I(X; T)$. Section 8 recasts the bridge as a Bayesian architecture-comparison rule. The ladder is an empirical realization of the comparison in that theorem: measured predictive gain equals representation gain net of the change in within-representation excess risk.

3.1 An architecture ablation on 701 economic time series

The top rung of the ladder is *laplace*, a new online distributional forecaster built as a likelihood-weighted ensemble over a population of composed transforms. Its algorithmic specification and baseline comparisons are documented with the library (Cotton, 2026). Here it serves as the measurement instrument, and the lower rungs are restrictions of the same design. We ran the nested ladder on 701 economic time series drawn from the FRED database by fixed enumeration rules rather than hand-curation. Each series enters as changes: first differences, or log differences when the levels are strictly positive. The choice is made once from the full series and is shared by every rung, so it does not differentiate them. The corpus is filtered to continuous series, meaning fewer than five percent repeated consecutive changes, with at least 600 observations. Forecasting is online and one step ahead.

The score is an exact log score. Every rung’s forecast is evaluated as the mixture $(1 - \delta) \hat{Q}_j + \delta G_0$, with $\delta = 0.01$ and G_0 the expanding-window Gaussian fitted to the raw changes. No clipping or truncation is applied, so the reported numbers are genuine log-score differences of proper forecasts. The mixture component is the best-scoring one on at most a third of one percent of observations at every rung.

The benchmark average weights series equally: each series contributes its per-observation mean, with the first 301 observations unscored as burn-in. FRED contains related and occasionally near-duplicate quantities, so the 701 series are not independent draws. The intervals reported below resample whole series and are descriptive, since cross-series correlation is not modeled.

Ablation player	Exact candidates the rung adds
location (rung 1)	exponential moving averages at rates 0.01, 0.05, 0.1, 0.3; first differencing; adaptive drift at four speed-shrinkage settings; the theta method at rates 0.05, 0.1, 0.3; Holt linear level-plus-trend at three settings; all with pooled (expanding-window) Gaussian residual leaves
scale (rung 2)	the same mean candidates re-equipped with EWMA conditional-variance Gaussian leaves, plus a bare EWMA-variance leaf
seasonal (rung 3)	seasonal differencing at periods 7, 12, 24 (both leaf types); the hedged seasonal anchor at the same periods; seasonal differencing composed with an exponential moving average
serial (rung 4)	autoregression of order one (both leaf types) and order two with decayed coefficients; fractional differencing at $d = 0.2, 0.4$ over an exponential moving average; differencing over exponential moving averages at three rates
final (rung 5)	everything in <i>laplace</i> not present at rungs 1–4, a bundle of shape, coordinate, and variance dynamics: the Gaussian-scale-mixture terminal leaf and likelihood-weighted terminal ensemble (learning rate 0.8, depth penalty 0.005, forgetting 0.99, lattice projection); EWMA standardization compositions, including fast-tracker-over-slow-scale at two timescales; GARCH(1, 1) over an exponential moving average; a square-root power transform; Yeo–Johnson at $\lambda = 0, 0.5$ over differencing or an exponential moving average

Table 1: The definitive definition of the ablation’s five players: rung j of the ladder adds exactly row j to the pool, and the attribution in Table 2 is with respect to these assignments. The final player is a bundle, and in particular GARCH-type variance dynamics arrives at rung 5, not with the EWMA scale family at rung 2, so the “shape” share includes variance dynamics beyond a single-speed EWMA. A mean-reversion (Ornstein–Uhlenbeck) group exists in the library but is active only at multi-step horizons and plays no role here.

Each rung is a likelihood-weighted ensemble over a candidate pool that strictly contains the previous rung’s pool:

Rung	Adds to the previous rung
T_0 unconditional	$N(0, \hat{\sigma}^2)$ on changes, expanding-window variance
T_1 +location	exponential-moving-average, drift, theta, Holt, and differencing mean trackers (standard smoothing devices), pooled residual law
T_2 +scale	EWMA conditional-variance residual leaves
T_3 +seasonal	seasonal differencing and seasonal anchors
T_4 +serial	AR(1), AR(2), fractional differencing
T_5 full grammar	coordinate transforms, GARCH, mixture shape

Table 1 defines the five ablation players exactly, as implemented.

This is an *architecture ablation*: successive rungs change the whole forecaster, including its mean dynamics, so the measured values price increasingly rich forecasting architectures under this benchmark. They do not isolate the residual-pooling floor of a conformal method built on a fixed point predictor: a global residual conformal method may use an arbitrarily sophisticated point forecast and still pool its residuals, so its location representation need not stop at the ladder’s location rung. The matched-predictor decomposition of Section 3.2, equation (6), is the

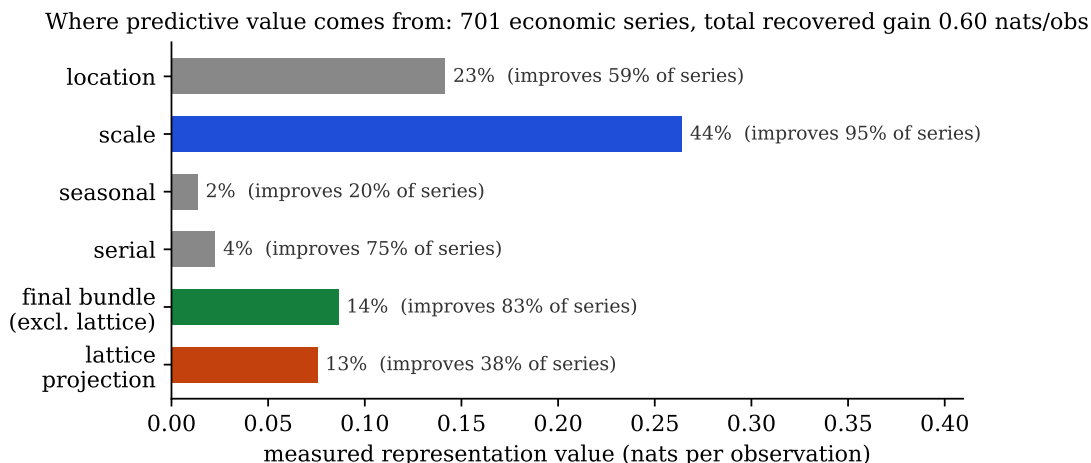


Figure 1: Measured representation values on 701 continuous FRED change series, under the nesting order declared in advance. Bar lengths are held-out one-step log-score gains in nats per observation from moving from each rung to the next architecture; the final-system increment is split into its lattice-projection component and the remainder. Annotations give each bar’s share of the mean total recovered gain and the fraction of series improved.

separate experiment that isolates it.

The final rung is the complete implemented forecaster, so it bundles architectural changes beyond new transform families. We quantify the largest such component, the lattice projection, in the sensitivity analysis below. Figure 1 and Table 2 report the measured representation values under the declared nesting order.

We highlight four features of the measurement that bear directly on the theory.

First, the full grammar recovered a positive predictive gain relative to the trivial architecture on every one of the 701 series, with mean 0.60 nats per observation and per-series quartiles 0.15, 0.31, and 0.65. By Proposition 2 this realized architecture gain equals the representation-floor reduction plus the change in endpoint within-representation excess risk. It shows that the richer grammar matters without identifying how much of the total is population information rather than endpoint fitting improvement. It is, at any rate, not a measure-zero curiosity on this benchmark.

Second, conditional scale is the largest and by far the most reliable family under the declared order: 44% of the mean gain, improving 95% of series. This is the piece a globally pooled residual law cannot represent and a scale-normalized law can. Read cumulatively: the location-only architecture trails the full grammar by a mean 0.46 nats per observation, and 57% of that shortfall is recovered by the scale family alone.

Third, location is bimodal where scale is uniform. Its median value is zero and it improves only 59% of series (many economic change series are close to white noise in the mean), but on the 28% of series where it is worth more than 0.01 nats, it is worth 0.50 nats on average. Averages conceal this composition, and the per-series distribution is part of the answer.

Fourth, the near-zero seasonal family illustrates that the measurement is relative to an empirical prior, and that prior should be named precisely: sufficiently long, continuous, mostly nonseasonal-or-adjusted FRED series, transformed to changes, scored one step ahead. On an hourly, retail, or load corpus the same family is known to be large. The measurement is a map from a benchmark distribution to representation values, $\Pi \mapsto \{\Delta_j\}$, and this corpus is

Rung	Δ_j mean	90% CI	median	share	improves
+location	+0.142	[0.114, 0.173]	0.000	23%	59% of series
+scale	+0.264	[0.228, 0.309]	+0.114	44%	95% of series
+seasonal	+0.014	[0.004, 0.026]	−0.000	2%	20% of series
+serial	+0.023	[0.019, 0.026]	+0.002	4%	75% of series
+final-system bundle	+0.162	[0.142, 0.185]	+0.050	27%	88% of series
total $T_0 \rightarrow T_5$	+0.604		+0.310		100% of series

Table 2: Measured representation values on 701 continuous FRED change series: held-out one-step log-score gains, in nats per observation, from adding each transform family to the nested pool. Intervals are series-resampling descriptive intervals: whole series are resampled, which respects dependence within a series but not correlation between related FRED series, so they should be read as descriptive rather than formal sampling intervals. “Share” is the rung’s mean value as a fraction of the mean trivial-to-full-grammar gain.

one point of that map. Running the frozen grammar on deliberately different corpora is the natural extension, and would connect directly to the prior-dependence of Theorem 13. The serial family makes the converse point: small in magnitude (4%) yet positive on 75% of series, cheap and consistently available. The final-system bundle is the second-largest component, at 27% and improving 88% of series. It bundles residual shape beyond a Gaussian with tracked scale, coordinate transforms, GARCH-type variance dynamics, and the implementation changes quantified below.

Sensitivity of the final-system increment

The family labels are also approximate at the edges: the location rung includes differencing trackers, and the final rung mixes residual shape, coordinate transformations, and GARCH-type variance dynamics, which is conditional-scale dynamics by another name. We therefore call the rung the *final-system bundle* rather than a transform family. The final rung also changes the architecture, not only the pool. It is the complete implemented forecaster, with a terminal-leaf ensemble in place of full-mixture ensembling, geometric forgetting, a tail splice, a lattice projection that places atoms on exactly repeating values, and plain-variance candidate leaves in place of rung 2’s EWMA-variance leaves. So Δ_5 is the marginal value of the complete forecaster over rung 4, a bundle of grammar and architecture. We measured the largest bundled component directly. The corpus filter excludes only consecutive repetition, and isolated exact repeats remain (the median series has 1% exact-zero changes), which the lattice projection rewards with sharp atoms. Rescoring the complete forecaster with the projection disabled splits Δ_5 : the lattice contributes a mean 0.076 nats (median zero, positive on 38% of series, concentrated on near-lattice series), and the remaining 0.087 nats (positive on 83%) is the grammar-and-architecture part. Without the lattice the total recovered gain is 0.53 nats, still positive on every series, and the scale family’s share rises to 50% (`r5_nosticky.py`). A Shapley analysis over the 2^5 family subsets gives an insertion-order-robust attribution for this fixed grouping. It is not robust to regrouping, and moving GARCH from the shape family to the scale family would change it. We computed exact Shapley values for an explicitly defined surrogate coalition game: for a coalition A of families, let $\hat{Q}_A$ be the likelihood-weighted pool built from exactly the candidates of the families in A , and set $v(A) = \hat{\mathcal{L}}(\hat{Q}_\emptyset) - \hat{\mathcal{L}}(\hat{Q}_A)$, the recovered gain over the empty pool, so that $v(\emptyset) = 0$, larger is

better, and the Shapley values sum to the full-coalition recovered gain.

The final family needs a compositional realization, since its production form presupposes the lower rungs: its mixture leaves use the EWMA scale when the scale family is present and a near-expanding scale otherwise. The game was evaluated on a deterministic third of the corpus (234 series, all 2^5 coalition pools, with total recovered gain 0.35 nats per observation, smaller than the main ladder’s because the surrogate final family is weaker than the complete forecaster). To isolate the order effect we also recompute the declared-order attribution on the same game and series:

Family	declared order (same game)	Shapley (same game)
location	42%	12%
scale	48%	42%
seasonal	1%	1%
serial	7%	19%
shape (final)	3%	26%

Scale is the largest family under both attributions, which is the headline robustness check. The order effects within the same game are otherwise large: the declared order, which inserts location first and the final family last, credits location with structure that Shapley reassigns to the serial and final families, since an autoregressive or shape state can mimic slow location tracking and conversely. Seasonal is negligible under both. These same-game shares are not directly comparable with Table 2, which uses the complete forecaster as the final rung on the full corpus.

Two disclosures protect the grammar-as-evidence reading. The grammar and its hyperparameters were developed against FRED-based benchmarks that overlap this corpus, so the corpus is not a holdout with respect to the grammar’s own selection history. The enumeration and filtering rules for the 701 series are fixed and reproducible, the nesting order was declared before the ablation was run, and the repository commit is recorded. As the out-of-development check, we reran the identical ladder on a separate holdout of 300 series: the least popular members of the same FRED daily universe, none ever fetched or used during the grammar’s development (`ladder_holdout.py`). The findings replicate. The total recovered gain is 0.34 nats per observation, positive on 99.7% of the holdout series. Conditional scale is again the largest and most reliable family: 50% of the gain, improving 98% of series (90% interval [0.14, 0.21] nats). Seasonal is again negligible, and the final family’s share is smaller, 13%, on these less structured series. The lattice split replicates as well: disabling the projection leaves a total of 0.32 nats, positive on 99.7%, with the scale share at 54%. The series are revised FRED values. Revised and real-time-vintage series are both legitimate objects of study, just different ones, and the findings here concern the former.

3.2 Transform, then pool, and conformalize last

A residual-based forecasting pipeline has three stages: model or transform; estimate or pool the residual law; conformalize, meaning apply the rank-based adjustment that supplies a finite-sample certificate. The first two stages determine the representation floor and the base predictive law. Conformal rank adjustment supplies a certificate and may also alter the post-processing component of predictive risk. It is priced separately in Section 10.

Methods that look philosophically opposed are then just different stopping rungs on one ladder. A full transform stack composes state-revealing transformations (location, scale, season, serial

state, coordinate) until what remains is as close to white as the grammar permits, and attaches a minimal residual law at the bottom. Signed-residual predictive systems pool the raw residuals immediately after the point forecast. Ordinary symmetric split-conformal regression pools their absolute values instead. Both stop before modeling conditional residual state. Normalized variants stop one rung later, and distributional recalibration pools at the top. The substantive question is not which philosophy to hold but whether homogenization has been carried far enough before pooling. And the early stop is tempting because the marginal coverage guarantee is indifferent to it. Coverage holds whether or not the residual stream is conditionally homogeneous, so marginal validity can coexist with substantial conditional heterogeneity, which is exactly what the information gap measures.

Stated this way, residual pooling is *early stopping*: every method chooses where to stop transforming and attach a residual-law estimator. As with early stopping in estimation, the stop can act as regularization. The empirical law at the stopping rung imposes no parametric shape restriction. Under the usual i.i.d. or stationary-ergodic conditions it converges to the pseudo-true law of whatever the retained transforms leave behind. The question the decomposition below makes precise is whether the regularization benefit of the empirical estimator outweighs the representation cost of stopping.

Validity and efficiency separate here, and the point deserves care. In the i.i.d. regression setting, residuals from a fixed point predictor are exchangeable even when $I(R; X) > 0$. Heteroscedastic residuals are still marginally i.i.d., and split conformal retains its marginal validity. The information gap is not a failure of exchangeability. It is the cost of conditional heterogeneity, $P_{R|X} \neq P_R$, an efficiency statement, not a validity statement.

What a transform stack manufactures is conditional homogeneity. It moves the residual stream toward coordinates in which the conditional law no longer varies with the state. The floor quantifies what remains: $I(R_t; \mathcal{F}_{t-1} | T_t)$ is the predictive information still flowing from the past into the rung- T_t residual stream. Transform until conditionally homogeneous, pool in those coordinates, and conformalize afterward when a certificate has value. On this reading, raw residual pooling stops before homogeneity has been manufactured, and the measured cost of the early stop is the price of the heterogeneity still in the stream.

For time series, exchangeability is a separate and additional issue: even $I(R_t; \mathcal{F}_{t-1} | T_t) = 0$ does not by itself make the transformed score sequence exchangeable without a time-invariant score law and appropriate serial-independence conditions.

This also relocates the critique of pooling. Every transformed-pooling construction considered here accumulates a single empirical distribution across time. The normalized and recalibrated variants are no exception: they pool standardized residuals and probability integral transforms respectively. Pooling as an estimator is robust on the evidence below, where it beats its fitted counterpart at every depth. Pooling in coordinates that leave the residual law dependent on state creates a positive irreducible floor (Theorem 11). On this benchmark the raw residual scale is such a coordinate system, while the coordinates in which the conditional law is homogeneous lie one or more transforms deeper. The two decisions separate cleanly. Whether to pool is an estimation trade-off between an empirical law and a fitted family, and either choice can win at a given sample size. What to pool fixes the coordinates, and with them the floor that no estimator of either kind can cross.

Empirical pooling does not extrapolate beyond observed extremes, so it says nothing about tails. This matters little under the quantile-grid score used below, but it can dominate the log score. Appendix A discusses the calibration-resolution limit and compares empirical, parametric,

and semiparametric terminal laws.

The depth-versus-estimator choice decomposes exactly. Fix the point predictor. At each rung T_j , compare a score-trained residual law $\hat{Q}_j$ with the empirical transformed-residual forecast $\hat{E}_j$, the empirical law of the T_j -transformed residuals. We write $\hat{E}_j$ rather than a conformal symbol deliberately: the empirical law is a legitimate nonparametric estimator in its own right, and the conformal rank construction that would supply a validity certificate is a separate operation applied on top of it. Relative to the richest fitted rung,

$$\hat{\mathcal{L}}(\hat{E}_j) - \hat{\mathcal{L}}(\hat{Q}_m) = \underbrace{\hat{\mathcal{L}}(\hat{E}_j) - \hat{\mathcal{L}}(\hat{Q}_j)}_{\text{empirical versus fitted at fixed depth}} + \underbrace{\sum_{k=j+1}^m [\hat{\mathcal{L}}(\hat{Q}_{k-1}) - \hat{\mathcal{L}}(\hat{Q}_k)]}_{\text{fitted-ladder gain forgone by stopping}}. \quad (6)$$

The first term compares the empirical estimator with the fitted family at a fixed representation. It mixes model-form restriction with finite-sample estimation differences, and its sign is an empirical matter. The second term is the fitted-ladder gain forgone by the early stop. The decomposition separates what residual methods bundle: how much of a method’s predictive shortfall is its estimator, and how much is its stopping depth.

There is a third axis besides depth and estimator: the scoring rule against which forecasts are ultimately evaluated. Chasing coverage optimizes a different functional than minimizing a proper score, even at the same rung with the same estimator. The reason is that a fixed-level interval criterion is proper only for the corresponding quantile pair, not for the predictive density (Section 5). A method tuned for fixed-level coverage and one trained on CRPS or log score will generally disagree, and each is optimal for its own objective. The comparisons below settle everything under proper scores. The value of the coverage guarantee itself is accounted separately in Section 10.

Matched-predictor experiment: design and scoring

We ran the depth-by-estimator grid on the same 701 series with a single fixed point predictor for every cell: the full grammar’s mean forecast, computed online. We call the residual law attached at the bottom of a transform stack its *terminal leaf*. The four residual depths, with their fitted and empirical laws, are:

Depth	Transformation	Fitted law $\hat{Q}_j$	Empirical law $\hat{E}_j$
T_0	raw residual	Gaussian	empirical residual law
T_1	EWMA scale normalization	scaled Gaussian	empirical standardized law
T_2	AR(1) whitening	Gaussian innovation law	empirical innovation law
T_3	distributional (PIT) transform	Gaussian scale mixture	empirical PIT recalibration

The quantiles of $\hat{E}_0$ give one-sided split-conformal bounds, $\hat{E}_1$ is the law underlying normalized conformal, and $\hat{E}_3$ underlies distributional recalibration. The usual fixed-level symmetric interval is evaluated directly in the simulation below. All cells use the same online point forecast. No conformal rank adjustment is applied, and under temporal dependence the ordinary split-conformal exchangeability guarantee would not cover these series in any case: this experiment compares residual-law estimators and representations, not finite-sample certificates.

We settle every cell with a 99-point quantile approximation to CRPS, the equally weighted average of pinball losses on the grid $0.01, \dots, 0.99$. It is proper for the reported quantile vector, defined for empirical laws, and identical for every cell. Each series is normalized by its own $\hat{Q}_0$

value before averaging. Since that denominator depends on realized outcomes, the cross-series relative summary is descriptive. Only the per-series criterion is asserted to be proper for the reported quantile vector. Because the grid stops at the first and 99th percentiles it mechanically mutes the extreme tails, and tail-related conclusions under this score should be read at that resolution.

rel. 99-ctl. CRPS	T_0 const	T_1 scale	T_2 +serial	T_3 +shape
fitted $\hat{Q}_j$	1.000 / 1.000	0.951 / 0.976	1.091 / 0.991	1.280 / 0.992
empirical $\hat{E}_j$	0.957 / 0.988	0.938 / 0.971	0.961 / 0.974	1.011 / 0.983

mean / median across 701 series; lower is better; best cell in bold.

We report four findings. First, the empirical estimator outperforms its fitted counterpart at every depth, winning on 81%, 78%, 97%, and 87% of series at depths 0–3 (paired series-resampling 90% intervals for the mean effect: $[-4.9, -3.7]$, $[-1.5, -1.1]$, $[-16.1, -10.2]$, and $[-40.1, -16.1]$ percent of the pooled score). The advantage is modest at the raw and scale-normalized depths and much larger where the fitted serial and shape models are unstable, with mean relative scores of 1.09 and 1.28 for the fitted serial and shape laws against stable empirical counterparts. The result is consistent with the projection interpretation, since the empirical estimator avoids the shape restrictions of the fitted families, though the asymptotic projection theorem does not itself imply finite-sample dominance. The pattern is consistent with empirical pooling acting as a useful regularizer at these sample sizes.

Second, the best cell on this benchmark is $\hat{E}_1$: scale-normalized empirical pooling, the predictive core of normalized conformal. It has the best mean and median and is the single best cell on 54% of series. The scale rung is best on 70% with either estimator, and an empirical cell is best on 81%.

Third, within this grid the main deficit of raw pooling is attributable to its coordinates rather than to empirical estimation. $\hat{E}_0$'s empirical-versus-fitted term is favorable, at a mean -4.3% of the pooled score. But it trails $\hat{E}_1$ by 2.0% on average (90% interval $[1.8, 2.2]$, median 1.4%).

With the point predictor held fixed, scale-normalized empirical pooling $\hat{E}_1$ beats raw empirical pooling $\hat{E}_0$ on 87% of the 701 series under the quantile-approximated CRPS.

This is the single number most directly relevant to the conformal thesis. The measured deficit of raw residual pooling arises from stopping before the scale state, not from empirical estimation. The serial and shape rungs add essentially nothing here. The result shows that, for this fixed predictor, these particular AR and shape refinements recover little additional value under the quantile-grid score. That may be because much mean dependence has already been removed, because the remaining dependence is weak, or because the tested transforms and score do not capture its value. The fitted columns use deliberately simple parametric families, so the comparison is between estimator classes at a rung, not a claim that no better parametric practice exists.

Fourth, the scoring rule chooses the terminal leaf. As a reference cell we scored the full grammar's own distributional forecast, the full transform stack with a fitted terminal residual law. Its relative score is 0.949 mean and 0.976 median. Under CRPS it is beaten by $\hat{E}_1$ on 80% of series, by a mean 1.1% of the pooled score (90% interval $[0.9, 1.3]$). Under the log score, the ladder of Section 3.1 shows the fitted mixture leaf well ahead of any scale-rung law. The

scoring rule shapes the preferred terminal estimator: pool the standardized residuals empirically under either score, model the tails when the score is log, and prefer a score-weighted ensemble of leaves overall (Appendix A). Component-level leaf comparisons did not, however, translate into an improvement when the alternatives were installed in the complete forecaster. The appendix shows that the reference system had already captured much of the apparent gain elsewhere in its architecture.

The certificate under valid conditions: a simulation

The real-data grid deliberately applies no conformal rank adjustment, so one small i.i.d. simulation completes the story in the exact setting where conformal validity applies. Let $X \sim \text{Unif}[0, 1]$ and $Y = \sigma(X)\varepsilon$ with $\sigma(x) = 0.5 + 2x$ and $\varepsilon \sim N(0, 1)$, the mean known to be zero. We estimate a scale proportional to σ by regressing $|Y|$ on X in a training split of 1000. The regression estimates $\sqrt{2/\pi}\sigma$, and the constant cancels in the normalization. Raw and normalized split conformal then use the same calibration set of 500 and the same rank index, at $\alpha = 0.1$. Each of 400 replications scores 20,000 test points, and Monte Carlo standard errors of all reported coverages are at most 0.001 (`check_conformal_sim.py`):

	marginal	σ -qrt. 1	qrt. 2	qrt. 3	qrt. 4	width	int. score
raw split conformal	0.901	0.998	0.964	0.873	0.767	5.37	7.26
normalized	0.901	0.900	0.901	0.901	0.900	4.96	6.21

Both methods achieve the target 90% marginal coverage to Monte Carlo precision. Raw pooling, however, produces severe state-dependent variation in coverage. It overcovers the calm quartile, covers only 77% of the most volatile, and pays in width and interval score. Normalized pooling has approximately uniform coverage across all four scale quartiles in this simulation. The order-statistic rule is identical in both rows, so the entire difference is the pooling coordinates: the certificate can remain valid while the coordinates remain inefficient.

The preferred terminal estimator depends on the scoring rule, and a bake-off of terminal leaves supporting that statement is in Appendix A.

Remaining predictability, measured against a pooled baseline

The floor language invites a direct check: how much conditional predictability actually remains in the transformed residual stream at each depth? For each series we form four streams: raw residuals, standardized, serially whitened, and PIT-normalized. On each stream we compare two procedures. The challenger is a fresh instance of the full-grammar forecaster run on the stream, so it sees the stream’s own past H . The pooled baseline is a kernel estimate over the stream’s past values. Both are floored by the same reference mixture, so each is a genuine density. Both procedures are measurable with respect to the stream’s past: the pooled kernel estimate is itself computed from H , so writing $Q_{\cdot|H}$ for the challenger and M_H for the pooled estimator, the reported quantity is exactly

$$\mathbb{E} \log \frac{q(Z|H)}{m_H(Z)} = \mathbb{E}_H \text{KL}(P_{Z|H} \| M_H) - \mathbb{E}_H \text{KL}(P_{Z|H} \| Q_{\cdot|H}),$$

the held-out difference between two H -measurable predictive procedures. It measures whether the structural challenger exploits the past more effectively than the expanding pooled estimator.

It contains no isolated mutual-information term, and it is neither an estimate of nor a bound on $I(Z; H)$. A genuine information lower bound would need a joint-versus-product variational objective of Donsker–Varadhan or NWJ type (`conformal_infogap.py`):

Stream	challenger gain (nats/obs)	90% interval	positive on
raw residuals	+0.145	[0.133, 0.157]	87% of series
standardized	+0.018	[−0.020, 0.051]	45%
whitened	+0.021	[−0.016, 0.055]	44%
PIT-normalized	+0.006	[−0.004, 0.015]	27%

The raw residual stream of even this strong mean forecast supports a substantial challenger gain, and a single scale standardization eliminates detectable challenger advantage relative to the pooled baseline, where the deeper transforms leave it. Read at its stated resolution, this is the homogenization claim: on this benchmark, one transform removes the conditional structure that this challenger can exploit against this baseline, consistent with the grid’s finding that the scale rung is where pooling becomes competitive. Zero certifies only that this particular challenger finds nothing against this particular pooled estimate. A negative value records challenger underperformance relative to the pooled estimator. It is neither negative information nor evidence that the stream is homogeneous.

4 The representation decomposition

The experiments leaned on a handful of identities, and we now state them with care. We begin with the value-of-information decomposition itself, then extend it to proper scoring rules and to learned representations. Section 6 then specializes the decomposition to residual systems, where the limitations of residual pooling take their exact form and Theorem 11 prices the coordinate choices the experiments measured.

Let $(X, Z) \sim P$. Assume that $P_{Z|X}$ and the candidate forecasts admit densities with respect to a common dominating measure and that the expectations below are finite. Let $T = T(X)$ be measurable.

Notation. For laws P and Q with densities p and q , $\text{KL}(P \parallel Q) = \mathbb{E}_P \log(p/q)$ is the Kullback–Leibler divergence, and $P \otimes Q$ denotes the product law under which the two coordinates are independent with marginals P and Q . The mutual information between Z and X is $I(Z; X) = \text{KL}(P_{Z,X} \parallel P_Z \otimes P_X)$, and the conditional mutual information given T is $I(Z; X | T) = \mathbb{E}_T \text{KL}(P_{Z,X|T} \parallel P_{Z|T} \otimes P_{X|T}) = \mathbb{E} \log [p(Z | X)/p(Z | T)]$, the second equality holding because T is a function of X . $\mathbb{E}_T$ denotes expectation over T , $H(Z | T)$ denotes conditional entropy, and $h(\cdot)$ differential entropy, both entropies in the sense matching the dominating measure. All logarithms are natural, so information is measured in nats.

We state the decomposition as a lemma because it is classical. The paper’s substantive results are its specializations in Sections 6–9.

Lemma 3 (Value of information under logarithmic loss). *For any conditional predictive density $q(z | T)$,*

$$\mathcal{R}(q; T) := \mathbb{E} \left[\log \frac{p(Z | X)}{q(Z | T)} \right] = I(Z; X | T) + \mathbb{E}_T \text{KL}(P_{Z|T} \parallel Q_{Z|T}). \quad (7)$$

Therefore

$$\inf_q \mathcal{R}(q; T) = I(Z; X | T),$$

with the infimum attained at $q(\cdot | T) = p(\cdot | T)$ almost surely.

Proof. Insert the true T -conditional density:

$$\log \frac{p(Z | X)}{q(Z | T)} = \log \frac{p(Z | X)}{p(Z | T)} + \log \frac{p(Z | T)}{q(Z | T)}.$$

Take expectations of each term. For the first, since $T = T(X)$,

$$\mathbb{E} \log \frac{p(Z | X)}{p(Z | T)} = I(Z; X | T)$$

by the definition in the notation paragraph. For the second, condition on T and use the tower property: given T , the outcome Z has density $p(\cdot | T)$, so

$$\mathbb{E} \log \frac{p(Z | T)}{q(Z | T)} = \mathbb{E}_T \mathbb{E} \left[\log \frac{p(Z | T)}{q(Z | T)} \middle| T \right] = \mathbb{E}_T \text{KL}(P_{Z|T} \| Q_{Z|T}).$$

Summing the two gives (7). The second term is nonnegative and vanishes exactly when $Q_{Z|T} = P_{Z|T}$ for P_T -almost every T , which yields the infimum and its attainment. $\square$

Remark 4 (Pseudo-truth). In the language of White (1982), $P_{Z|T}$ is the pseudo-true member of the T -measurable class, and $I(Z; X | T)$ is the specification error of the class itself. An estimator can be consistent for the pseudo-truth, so that the second term vanishes, while the first term remains.

Corollary 5 (Value of a representation). *For deterministic $T = T(X)$,*

$$I(Z; X) = I(Z; T) + I(Z; X | T).$$

Thus the reduction in the oracle floor from using T rather than a constant representation is $I(Z; T)$.

Corollary 6 (Nested representations). *If $T_A = g(T_B)$, then*

$$I(Z; X | T_A) - I(Z; X | T_B) = I(Z; T_B | T_A) \geq 0.$$

Remark 7 (Partial rather than total order). A scale estimate, a partition, a neighborhood, and a learned embedding need not be comparable. Monotonicity applies under deterministic refinement or, more generally, an appropriate Blackwell ordering. A long list of methods should therefore be regarded as a partially ordered catalogue, not automatically as a single ladder.

4.1 Learned representations

Let D_n be training data independent of the test pair conditional on the data-generating law. Let $T_n = T_n(X; D_n)$ and let $\hat{q}_n(\cdot | T_n, D_n)$ be learned from D_n . Conditional on D_n , Lemma 3 applies to the random architecture realized by the training sample:

$$\begin{aligned} \mathcal{R}_n(\hat{q}_n | D_n) &:= \mathbb{E} \left[\log \frac{p(Z | X, D_n)}{\hat{q}_n(Z | T_n, D_n)} \middle| D_n \right] \\ &= I(Z; X | T_n, D_n) + \mathbb{E} \left[\text{KL}(P_{Z|T_n, D_n} \| \hat{Q}_{n, \cdot | T_n, D_n}) \middle| D_n \right]. \end{aligned} \tag{8}$$

The unconditional risk is obtained by averaging (8) over D_n . This conditioning is necessary whenever the representation, the point forecast, or the residual transformation is learned.

5 Proper scoring rules

Let $S(Q, z)$ be a negatively oriented proper scoring rule on a class $\mathcal{P}$ of predictive distributions, with generalized entropy

$$H_S(P) = \mathbb{E}_{Z \sim P} S(P, Z)$$

and score divergence

$$d_S(P, Q) = \mathbb{E}_{Z \sim P} [S(Q, Z) - S(P, Z)] \geq 0.$$

Assume all terms are finite and the relevant conditional laws belong to $\mathcal{P}$ (Gneiting and Raftery, 2007; Grünwald and Dawid, 2004).

Theorem 8 (Proper-score decomposition). *For any T -measurable forecast $Q_{\cdot|T}$,*

$$\mathbb{E}S(Q_{\cdot|T}, Z) - \mathbb{E}S(P_{Z|X}, Z) = \underbrace{\mathbb{E}_T H_S(P_{Z|T}) - \mathbb{E}_X H_S(P_{Z|X})}_{\mathcal{I}_S(Z; X | T)} + \mathbb{E}_T d_S(P_{Z|T}, Q_{\cdot|T}). \quad (9)$$

An optimal T -measurable forecast is $P_{Z|T}$, unique modulo null sets when S is strictly proper on the relevant class, and its irreducible score regret is $\mathcal{I}_S(Z; X | T)$.

Proof. Insert the optimal T -measurable forecast $P_{Z|T}$:

$$\mathbb{E}S(Q_{\cdot|T}, Z) - \mathbb{E}S(P_{Z|X}, Z) = \underbrace{\mathbb{E}[S(Q_{\cdot|T}, Z) - S(P_{Z|T}, Z)]}_{(a)} + \underbrace{\mathbb{E}[S(P_{Z|T}, Z) - S(P_{Z|X}, Z)]}_{(b)}.$$

For (a), condition on T : given T , the outcome Z has law $P_{Z|T}$, so

$$(a) = \mathbb{E}_T \mathbb{E}[S(Q_{\cdot|T}, Z) - S(P_{Z|T}, Z) | T] = \mathbb{E}_T d_S(P_{Z|T}, Q_{\cdot|T}) \geq 0$$

by propriety. For (b), condition on T in the first summand and on X in the second:

$$\mathbb{E}S(P_{Z|T}, Z) = \mathbb{E}_T H_S(P_{Z|T}), \quad \mathbb{E}S(P_{Z|X}, Z) = \mathbb{E}_X H_S(P_{Z|X}),$$

so (b) equals the displayed entropy difference $\mathcal{I}_S(Z; X | T)$. It is nonnegative because, for each x , propriety of S under $P_{Z|X=x}$ gives $\mathbb{E}_{Z \sim P_{Z|X=x}} S(P_{Z|T(x)}, Z) \geq H_S(P_{Z|X=x})$, and averaging over X preserves the inequality. $\square$

The generalized information $\mathcal{I}_S$ is nonnegative by propriety. For log loss it is ordinary conditional mutual information. For the continuous ranked probability score (CRPS) it is a CRPS-specific entropy reduction, under finite-first-moment conditions. A single fixed-level interval score is proper for the corresponding pair of quantiles, not strictly proper for the entire predictive distribution. The Shannon chain rule, likelihood-ratio interpretation, and Kelly-betting interpretation used below are specific to log loss and should not be transferred automatically to arbitrary scores.

Sufficiency of a representation is likewise loss-dependent. Under log loss or any strictly proper distributional score, the Bayes action is the full conditional distribution, so sufficiency requires $P_{Z|T} = P_{Z|X}$. Under squared loss only $\mathbb{E}[Z | T] = \mathbb{E}[Z | X]$ is required, and the representation cost is $\mathbb{E}[(\mathbb{E}[Z | X] - \mathbb{E}[Z | T])^2]$. Under a fixed quantile loss only the relevant conditional quantile must be retained. The floor $\mathcal{I}(Z; X | T)$ is the correct price for full-distribution prediction under log loss, and $\mathcal{I}_S(Z; X | T)$ is its analogue for the score actually used. A representation can be badly insufficient for one objective and exactly sufficient for another.

6 Residual predictive systems

Fix a location predictor $\hat{\mu}$. It may be learned, but throughout this section it is treated as fixed or we condition on the data used to learn it. Define

$$R = Y - \hat{\mu}(X).$$

Let $r(r|x)$ be the true conditional residual density and $\bar{r}(r)$ its marginal density. A *single-shape conditional model* chooses one density h , with law H , and reports

$$q_h(y|x) = h(y - \hat{\mu}(x)).$$

Here $T = \text{const}$ refers to the residual-shape component, not to the full forecast for Y , which still depends on x through $\hat{\mu}(x)$.

Theorem 9 (Information projection of a single-shape residual system). *For any input-blind residual density h ,*

$$\mathcal{R}(h) = \mathbb{E} \log \frac{r(R|X)}{h(R)} = I(R; X) + \text{KL}(P_R \| H). \quad (10)$$

The pseudo-true member of the single-shape class is $H = P_R$, and the specification error of the class is $I(R; X)$.

Proof. Insert the marginal residual density $\bar{r}$:

$$\log \frac{r(R|X)}{h(R)} = \log \frac{r(R|X)}{\bar{r}(R)} + \log \frac{\bar{r}(R)}{h(R)}.$$

The expectation of the first term is $\mathbb{E} \log[r(R|X)/\bar{r}(R)] = I(R; X)$, the mutual information written with densities. The expectation of the second is $\mathbb{E} \log[\bar{r}(R)/h(R)] = \text{KL}(P_R \| H)$, since $R \sim P_R$ with density $\bar{r}$. Equivalently, this is Lemma 3 with $Z = R$ and $T = \text{const}$, in which case $I(R; X|T) = I(R; X)$ and the conditional KL term reduces to $\text{KL}(P_R \| H)$. $\square$

The information gap of residual pooling, $I(R; X)$, has five equivalent readings, each a short identity.

1. *False-independence cost.* By definition of mutual information, $I(R; X) = \text{KL}(P_{X,R} \| P_X \otimes P_R)$: the divergence of the joint law from the nearest world in which $R \perp X$ holds with the same marginals.
2. *Expected log likelihood ratio.* Writing the same definition with densities, $I(R; X) = \mathbb{E} \log [r(R|X)/\bar{r}(R)]$: the mean per-observation log likelihood ratio by which the conditional residual model beats the pooled model.
3. *Information projection.* For any input-blind law H with density h , inserting $\bar{r}$ into the ratio gives $\text{KL}(P_{X,R} \| P_X \otimes H) = I(R; X) + \text{KL}(P_R \| H)$, which is (10) in joint form. Minimizing over H yields $\inf_H \text{KL}(P_{X,R} \| P_X \otimes H) = I(R; X)$, attained at $H = P_R$. The gap is the distance from the joint law to the product family, and no input-blind step can cross it: the missing information is dependence, not marginal shape.

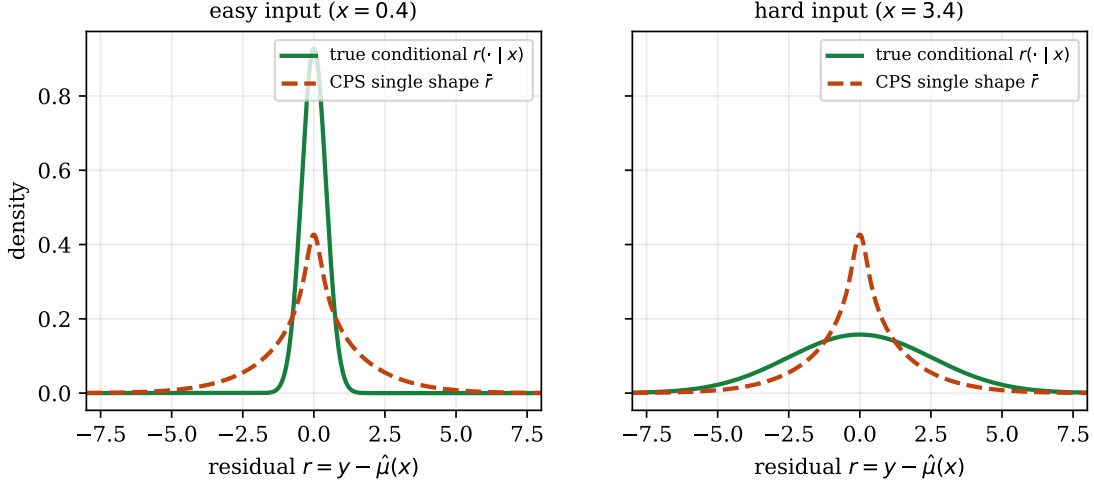


Figure 2: Simulated illustration of Theorem 9. A single input-blind residual law serves every input, while the conditional residual law is narrow where the input is easy and wide where it is hard. The log-score regret of the pooled system is $I(R; X)$.

4. *Conditional rank nonuniformity.* If the marginal residual CDF G is continuous and strictly increasing, then $U = G(R)$ is marginally uniform, the map is invertible so no information is lost, and $I(R; X) = I(U; X) = \mathbb{E}_X \text{KL}(P_{U|X} \parallel \text{Unif}[0, 1])$ (Section 6.4): the pooled ranks are uniform on average, and the gap is the average divergence of their conditional law from uniformity. Easy inputs concentrate U near one half, hard inputs push it into the tails, and a single pooled shape cannot tell them apart (Figure 2).
5. *Kelly betting rent.* Fix a reference (market) density m and let a bettor who allocates wealth by a density q receive payoff multiplier $q(R)/m(R)$. The expected log growth is $\mathbb{E} \log q(R) - \mathbb{E} \log m(R)$, and the second term does not depend on q . The best input-blind bettor therefore uses $\bar{r}$, and the conditional bettor's growth-rate advantage over it is $\mathbb{E} \log r(R|X) - \mathbb{E} \log \bar{r}(R) = I(R; X)$ (Kelly, 1956; Barron and Cover, 1988).

Items 1–3 restate the definition and Theorem 9. Item 4 needs the stated regularity, with a randomized PIT when atoms are present. Item 5 is specific to log score. Figure 2 illustrates the decomposition on a heteroscedastic example.

6.1 Signed conformal predictive systems

A signed-residual conformal predictive system based on an independent calibration sample estimates the full marginal residual CDF, with the usual finite-sample randomization details (Vovk et al., 2019). Its $1 - \alpha$ quantile, rank-adjusted to $\lceil (n+1)(1-\alpha) \rceil$, is the corresponding one-sided split-conformal bound. Central intervals can be formed from two signed-residual quantiles (Angelopoulos et al., 2024a). The usual symmetric two-sided split-conformal interval instead pools absolute residuals, and it is treated in Section 6.2.

Under continuity and standard convergence conditions, the empirical residual CDF converges to the marginal residual CDF G . Hence its large-calibration predictive law converges to

$$\Pi_x(y) = G(y - \hat{\mu}(x)).$$

Corollary 10 (Residual CPS information floor). *Under the convergence conditions above, the population limit of the signed-residual CPS is the marginal residual law P_R , the restricted optimum of Theorem 9. If P_R admits the required density, that limiting law has log-score regret $I(R; X)$ relative to the conditional residual oracle.*

In misspecification language: conformal calibration consistently estimates the pseudo-true member of a misspecified single-shape model, and consistency for the pseudo-truth does not remove the model’s specification error. This identifies the limiting law. It does not assert that the finite-sample CPS log risk converges. That would require a smoothed predictive density and conditions such as KL consistency, since empirical-CDF convergence alone does not control log score and the finite empirical law is discrete.

6.2 Absolute-residual intervals and symmetrization

In the population, or large-calibration, limit: if the nested family of ordinary absolute-residual split-conformal intervals is interpreted as a predictive distribution by sweeping the nominal level, the corresponding residual law is the symmetrization

$$h_{\text{sym}}(r) = \frac{1}{2} \{ \bar{r}(r) + \bar{r}(-r) \}.$$

Its log-score regret is

$$I(R; X) + \text{KL}(P_R \parallel H_{\text{sym}}).$$

The second term vanishes when the marginal residual law is symmetric. The symmetrization map can be interpreted as mixing P_R with its reflection. The KL penalty is bounded above by $\log 2$ whenever the relevant absolute-continuity conditions hold.

6.3 Coverage alone does not identify density quality

A set-coverage guarantee does not by itself constrain the quality of a separately attached density. Let $Y \sim N(0, 1)$ and use the fixed score $s(y) = |y|$ to produce a split-conformal interval. Attach instead the density $N(0, \sigma^2)$. The conformal interval and its coverage are unchanged as σ varies, while expected negative log score diverges to $+\infty$ as $\sigma \downarrow 0$ or $\sigma \uparrow \infty$. The observation is deliberately logical rather than algorithmic: it shows that a set certificate cannot serve as evidence for the quality of a density not used to construct the certificate (Recht, 2024). It does not claim that conformal predictive distributions themselves can be varied arbitrarily while holding their construction fixed.

6.4 Conditional ranks

Assume the marginal residual CDF G is continuous and strictly increasing on the support, and let $U = G(R)$. Then U is marginally uniform and the transformation is information preserving, so

$$I(R; X) = I(U; X) = \mathbb{E}_X \text{KL}(P_{U|X} \parallel \text{Unif}[0, 1]).$$

With atoms one should use a randomized PIT. Without invertibility, data processing gives only $I(U; X) \leq I(R; X)$. Quantitative conditional-uniformity diagnostics can therefore provide variational or classifier-based lower bounds on the representation floor. A hypothesis test alone does not automatically estimate such a bound.

6.5 Pooling in transformed coordinates

The empirical constructions of Section 3.2 pool residuals after a state-dependent change of coordinates. The gap of any such scheme has a closed form.

Theorem 11 (Transform pooling). *For P_X -almost every x , assume:*

1. $\phi_x : \mathbb{R} \rightarrow \mathcal{Z}$ is a strictly increasing C^1 diffeomorphism onto a common open interval $\mathcal{Z} \subseteq \mathbb{R}$;
2. $(x, r) \mapsto \phi_x(r)$ and $(x, r) \mapsto \partial_r \phi_x(r)$ are jointly measurable.

Let $Z = \phi_X(R)$, and let h be a density on $\mathcal{Z}$ with law H . A single transformed-shape model fixes one such h and reports

$$q_h(r | x) = h(\phi_x(r)) \left| \frac{\partial \phi_x(r)}{\partial r} \right|.$$

Then

$$\mathbb{E} \log \frac{p(R | X)}{q_h(R | X)} = I(Z; X) + \text{KL}(P_Z \| H). \quad (11)$$

The pseudo-true choice is $H = P_Z$, and the information gap of pooling in the coordinates ϕ is $I(Z; X)$.

Proof. Under the stated assumptions the change-of-variables formula gives, for each x ,

$$p(r | x) = p_Z(\phi_x(r) | x) \left| \frac{\partial \phi_x(r)}{\partial r} \right|,$$

where $p_Z(\cdot | x)$ is the conditional density of $Z = \phi_X(R)$ given $X = x$. Dividing by the definition of q_h , the Jacobians cancel:

$$\frac{p(R | X)}{q_h(R | X)} = \frac{p_Z(Z | X) |\partial_r \phi_X(R)|}{h(Z) |\partial_r \phi_X(R)|} = \frac{p_Z(Z | X)}{h(Z)}.$$

Now insert the marginal density p_Z of Z :

$$\mathbb{E} \log \frac{p_Z(Z | X)}{h(Z)} = \mathbb{E} \log \frac{p_Z(Z | X)}{p_Z(Z)} + \mathbb{E} \log \frac{p_Z(Z)}{h(Z)} = I(Z; X) + \text{KL}(P_Z \| H),$$

which is Theorem 9 applied to Z . □

Remark 12 (Learned transformations). If ϕ is learned from external training data D , independent of the test pair, apply the theorem conditionally on D : the identity reads, as in Section 4.1, $\mathbb{E}[\cdot | D] = I(Z; X | D) + \text{KL}(P_{Z|D} \| H_D)$ with $Z = \phi_{X,D}(R)$. For an online transformation whose state is generated from the current preforecast filtration, include that state in X or in T . Do not condition on the entire current history, which would fix the transformation and condition away the randomness being priced.

The theorem prices each pooling scheme by its coordinates: raw residual pooling has gap $I(R; X)$. Scale-normalized pooling has gap $I(R/\hat{\sigma}(X); X)$. Pooling of serially whitened residuals (the innovations Z_{white} left after an autoregressive fit) has gap $I(Z_{\text{white}}; X)$. Recalibration through a fitted conditional CDF $\hat{F}_x$ has gap $I(U; X)$ with $U = \hat{F}_X(Y)$, exactly when each $\hat{F}_x$ is strictly increasing and smooth onto a common range. Discrete or noninvertible fitted CDFs would need a separate randomized-transform formulation, since a randomized PIT is a stochastic kernel rather than a diffeomorphism. Absent such a formulation, data processing gives only the corresponding information bound. It also exposes a distinction that the next section's catalogue can blur: an architecture that estimates the full conditional law $P_{R|\hat{\sigma}}$ has floor $I(R; X | \hat{\sigma})$, whereas normalized pooling imposes the stronger restriction that $R/\hat{\sigma}(X)$ has one common shape, and its gap $I(R/\hat{\sigma}(X); X)$ is in general different. They are not the same architecture.

7 Representation-conditioned conformal methods

Let $T = T(X)$ be a representation used to stratify or normalize residuals. If an idealized method estimates the full conditional residual law $P_{R|T}$, then its oracle floor is $I(R; X | T)$. Examples include a finite partition, a scale state, or a learned embedding. These examples are generally only partially ordered.

Architecture	Residual representation	Oracle log-score floor
Global residual pooling	constant	$I(R; X)$
Group-wise (Mondrian) pooling	$B(X)$	$I(R; X B)$
Scale-conditioned residual model	$\hat{\sigma}(X)$	$I(R; X \hat{\sigma})$
Learned residual state	$T_\theta(X)$	$I(R; X T_\theta)$
Full conditional residual model	sufficient T ; in particular X	0

The table's floors price architectures that estimate the full conditional residual law given the stated representation. Pooled-shape schemes are priced instead by Theorem 11. Conformalized quantile regression and methods based on non-residual conformity scores fit most cleanly into the general (Z, T) theorem rather than automatically into the signed-residual notation (Romano et al., 2019; Chernozhukov et al., 2021). Their retained object may be a vector of fitted quantiles and their calibrated target may be a scalar nonconformity score. Any claim of an $I(R; X | T)$ floor should specify the residual or score variable being predicted.

Conformalization at a fixed representation can alter $Q_{R|T}$ and hence the estimation term. It may improve or worsen a proper score. It cannot change the lower bound $I(R; X | T)$ unless the conformity construction itself changes T .

8 Bayesian comparison of forecasting architectures

Architecture comparisons depend on how plausible data-generating laws are weighted. Let Nature draw a data-generating law $P \sim \Pi$, and let D_n be observed training data. Under posterior predictive logarithmic loss, define the Bayes-optimal predictive law conditional on a representation T by $P_\Pi(Y \in \cdot | T, D_n)$. For a T -measurable forecast $\hat{Q}$, write $\mathcal{L}_\Pi(\hat{Q} | D_n) = -\mathbb{E}_\Pi[\log \hat{Q}(Y | T, D_n) | D_n]$ for its posterior-predictive log risk and $\mathcal{L}_\Pi^*(T | D_n)$ for the risk of the Bayes-optimal choice. Information quantities subscripted by Π are computed under the joint law induced by Π given D_n .

Theorem 13 (Bayesian architecture comparison). *Let $T_A = g(T_B)$, so T_B refines T_A . Conditional on D_n , the difference between the oracle posterior-predictive log risks is*

$$\mathcal{L}_\Pi^*(T_A | D_n) - \mathcal{L}_\Pi^*(T_B | D_n) = I_\Pi(Y; T_B | T_A, D_n). \quad (12)$$

For fitted forecasts $\hat{Q}_A$ and $\hat{Q}_B$,

$$\mathcal{L}_\Pi(\hat{Q}_A | D_n) - \mathcal{L}_\Pi(\hat{Q}_B | D_n) = I_\Pi(Y; T_B | T_A, D_n) + \epsilon_A(D_n) - \epsilon_B(D_n), \quad (13)$$

where $\epsilon_j(D_n) = \mathbb{E}_\Pi \text{KL}(P_\Pi(Y | T_j, D_n) \| \hat{Q}_j(Y | T_j, D_n))$. Hence the richer architecture wins exactly when its added predictive information exceeds the change in its within-representation excess risk.

Proof. Work under the joint law induced by Π , conditionally on D_n . The Bayes-optimal T_j -measurable predictive law is the posterior predictive $P_\Pi(Y \in \cdot | T_j, D_n)$, and its log risk is the conditional entropy:

$$\mathcal{L}_\Pi^*(T_j | D_n) = -\mathbb{E}_\Pi[\log p_\Pi(Y | T_j, D_n) | D_n] = H_\Pi(Y | T_j, D_n).$$

Subtracting the two oracle risks and using $T_A = g(T_B)$, so that conditioning on T_B is the same as conditioning on (T_A, T_B) ,

$$\mathcal{L}_\Pi^*(T_A | D_n) - \mathcal{L}_\Pi^*(T_B | D_n) = H_\Pi(Y | T_A, D_n) - H_\Pi(Y | T_B, T_A, D_n) = I_\Pi(Y; T_B | T_A, D_n),$$

which is (12). For fitted forecasts, Lemma 3 applied at each T_j under this law gives $\mathcal{L}_\Pi(\hat{Q}_j | D_n) = H_\Pi(Y | T_j, D_n) + \epsilon_j(D_n)$. Subtracting the two fitted risks and substituting (12) gives (13). $\square$

Equation (13) gives the finite-sample architecture comparison at a fixed prior and sample size. A coarse architecture has a larger information floor but may be cheap to estimate. A richer architecture lowers the floor but may have higher finite-sample error. No universal ordering exists without specifying the prior, sample size, and score. Proposition 1 is the rung-by-rung specialization of (13), with the benchmark distribution in the role of Π . The ladder of Section 3 is an empirical realization of the full comparison: representation gain net of the change in within-representation excess risk. Figure 1 is that comparison at one point of the map $\Pi \mapsto \{\Delta_j\}$.

8.1 A two-regime prior favoring a tree-conditional learner

Let

$$X \sim \text{Bernoulli}(1/2), \quad Y | X=0 \sim N(0, 1), \quad Y | X=1 \sim N(0, a^2),$$

with $a \neq 1$. The conditional mean is zero in both regimes. Thus a global residual predictor and a conditional learner can share the same perfect point prediction.

The optimal pooled residual law is

$$M_a = \frac{1}{2}N(0, 1) + \frac{1}{2}N(0, a^2).$$

A depth-one conditional tree can represent the two regime-specific laws. Its oracle log-score advantage is

$$\begin{aligned} \Delta(a) &= \frac{1}{2}\text{KL}(N(0, 1) \| M_a) + \frac{1}{2}\text{KL}(N(0, a^2) \| M_a) \\ &= I(Y; X) > 0. \end{aligned} \tag{14}$$

It equals the Jensen–Shannon divergence between the two conditional laws and is bounded above by $\log 2$. As the regimes become nearly distinguishable from the observation, $\Delta(a)$ approaches one bit per observation.

Now let the two variances be unknown. For every fixed true $a \neq 1$, if the conditional learner’s excess KL risk vanishes and the pooled learner converges to M_a , then

$$\lim_{n \rightarrow \infty} \{\mathcal{L}_a(\hat{Q}_A) - \mathcal{L}_a(\hat{Q}_B)\} = \Delta(a) > 0.$$

Under a regular prior on the variances the same holds in prior-average risk with limit $\mathbb{E}_\Pi[\Delta(A)]$, given posterior consistency and enough integrability to exchange the limit with the prior average. At finite n , the conditional learner wins once the relevant Δ exceeds its extra within-representation

excess risk. This is a concrete theoretical reason that boosted distributional trees, quantile boosting, NGBoost, or separate boosted mean-and-scale models can outperform global residual conformal prediction under priors assigning substantial mass to piecewise-learnable conditional heteroscedasticity (Friedman, 2001; Chen and Guestrin, 2016; Duan et al., 2020).

The comparison is between prediction devices under the same score. Conformalization can be attached to the richer learner, but a global residual-pooling output remains a coarser predictive architecture than a model whose conditional scale or shape varies with X .

9 Time series and retained state

Let (Y_t) be a process and let $\mathcal{F}_{t-1}$ be the information available before predicting Y_t . Let $\hat{\mu}_t$ be $\mathcal{F}_{t-1}$ -measurable and define $R_t = Y_t - \hat{\mu}_t$. An architecture retains a state

$$T_t = T(\mathcal{F}_{t-1}).$$

The one-step log-score decomposition is

$$\mathbb{E} \log \frac{p(R_t | \mathcal{F}_{t-1})}{q(R_t | T_t)} = \mathbb{I}(R_t; \mathcal{F}_{t-1} | T_t) + \mathbb{E} \text{KL}(P_{R_t | T_t} \| Q_{R_t | T_t}). \quad (15)$$

Thus $\mathbb{I}(R_t; \mathcal{F}_{t-1} | T_t)$ is the predictive information in the past omitted by the retained state.

If the process is stationary, the relevant differential entropies are finite, and $\mathcal{F}_{t-1}$ is generated by the residual past, then the pooled-state floor is

$$\mathbb{I}(R_0; R_{-1}, R_{-2}, \dots) = h(R_0) - h(R_0 | R_{-1}, R_{-2}, \dots).$$

This is the gap between marginal residual entropy and the one-step conditional entropy rate. If the filtration includes exogenous variables or other information, the floor can be larger than the residual-past entropy-rate gap.

If the state map or its hyperparameters are learned from external training data D , apply (15) conditionally on D , or on a fixed initialization, with $T_t = T_D(\mathcal{F}_{t-1})$. The current preforecast filtration $\mathcal{F}_{t-1}$ remains random. Conditioning on the full current history would fix T_t and condition away exactly the randomness being measured. In nonstationary settings the one-step conditional decomposition remains meaningful, but the stationary entropy-rate identity need not hold. A linear-Gaussian state-space model is the extreme of representation efficiency: the filtering posterior is a finite-dimensional sufficient statistic of the past (Kalman, 1960), so its floor is zero, given known parameters and correctly specified dynamics. Under parameter uncertainty the sufficient state must also carry the posterior over parameters.

9.1 Volatility as an information state

Let returns satisfy

$$Y_t = \sigma_t \varepsilon_t, \quad \varepsilon_t \stackrel{\text{iid}}{\sim} N(0, 1),$$

where $\sigma_t > 0$ is $\mathcal{F}_{t-1}$ -measurable, ε_t is independent of $\mathcal{F}_{t-1}$, $\mathbb{E}|\log \sigma_t| < \infty$, and the marginal entropy below is finite. The conditional law is $N(0, \sigma_t^2)$. Global pooling sees the scale mixture $\bar{P}$. Its representation floor is

$$\mathbb{I}(Y_t; \mathcal{F}_{t-1}) = h(\bar{P}) - \frac{1}{2} \log(2\pi e) - \mathbb{E} \log \sigma_t.$$

A volatility state $\hat{\sigma}_t$ has floor $I(Y_t; \mathcal{F}_{t-1} | \hat{\sigma}_t)$. This floor is zero precisely when the retained state determines the true conditional scale, i.e. when σ_t is measurable with respect to $\hat{\sigma}_t$. Equality $\hat{\sigma}_t = \sigma_t$ is sufficient but not necessary.

The practical point is concrete here. Heavy tails in the pooled return distribution can arise because a single law mixes easy and hard volatility states. A volatility model reduces the representation floor by distinguishing those states. A conformal normalization may then certify quantiles or sets at the retained-state resolution, but the predictive gain comes from the volatility state and the fit, not from the coverage statement by itself.

10 Certification and conditional impossibility

Coverage is a property of sets or events, not a proper score for a complete predictive density. A conformal certificate can be extremely valuable when the decision problem explicitly rewards control of a miss rate, false-positive rate, or risk functional (Vovk et al., 2005; Angelopoulos et al., 2021; Bates et al., 2021; Angelopoulos et al., 2024b). It should nevertheless be accounted for separately from representation and estimation.

For a conformalized predictor with representation T , a useful ledger is:

$$\text{predictive regret} = \text{representation regret} + \text{estimation/post-processing regret},$$

plus a separate statement describing the coverage or risk certificate. The certificate may constrain or change the second term. It does not alter the first unless the method changes T .

Conditional-coverage no-go theorems state that exact, nontrivial, distribution-free conditional coverage cannot be guaranteed at arbitrary resolution over rich distribution classes (Lei and Wasserman, 2014; Foygel Barber et al., 2021). The information theorem states that coarse representations leave predictive information unused. Together they suggest a design tension, but not a one-dimensional identity. Fine certification difficulty depends not only on $I(Y; X | T)$ but also on the complexity and mass structure of T , the class of protected subgroups, and the assumptions on the data-generating process.

A more complete methodological objective would minimize something of the form

$$\underbrace{I(Y; X | T)}_{\text{oracle representation cost}} + \underbrace{\text{WithinRepErr}(T, \mathcal{Q}, n)}_{\text{model-form and fitting cost}} + \underbrace{\lambda \text{CertCost}(T, n, \alpha)}_{\text{certification cost}},$$

where $I(Y; X | T)$ becomes $\mathcal{I}_S(Y; X | T)$ when the scoring rule is not log score, and the last term is the statistical cost of providing the certificate at level α . Any decision value the certificate delivers belongs on the benefit side of the underlying decision problem, not in this risk sum. The weight λ is decision-specific. The optimal T depends on the prior on Nature, the sample size, the evaluation loss, and the utility assigned to certification.

The objective $\inf_T \{I(Y; X | T) + \lambda C(T)\}$ is a family of model-selection criteria indexed by the complexity penalty. Taking $C(T) = I(X; T)$ recovers the information bottleneck (Tishby et al., 1999). Dimension, partition cardinality, VC-type complexity, or expected excess risk give criteria in the spirit of penalized likelihood (Akaike, 1974). Finite-sample estimation and certification cost need not coincide with $I(X; T)$, which is why we keep the penalty abstract.

11 Asymptotic suboptimality of pooled residual architectures

Is coverage-driven residual prediction a dead end? As a universal statement, no. As a complete probabilistic forecasting architecture, global residual pooling can be asymptotically dominated under broad structured priors.

Suppose a prior Π concentrates on problems with learnable conditional residual structure and satisfies

$$\mathbb{E}_{\Pi} \mathbb{I}(R; X) \geq c > 0, \quad \mathbb{E}_{\Pi} \mathbb{I}(R; X | T_B) \leq d,$$

where T_B is the richer architecture’s retained representation, and let both methods share one fixed point predictor, so that R and hence $\mathbb{I}(R; X)$ do not depend on the method or the sample size. Define the prior-average risk advantage $A_n := \mathcal{L}_{\Pi, n}(\hat{Q}_{\text{pooled}}) - \mathcal{L}_{\Pi, n}(\hat{Q}_{\text{rich}})$. Suppose both learners converge in prior-average log risk to their architecture-specific pseudo-truths. The pooled learner’s limiting risk exceeds the conditional oracle’s by $\mathbb{E}_{\Pi} \mathbb{I}(R; X)$ and the richer learner’s by $\mathbb{E}_{\Pi} \mathbb{I}(R; X | T_B)$, so the advantage converges:

$$A_n \longrightarrow \mathbb{E}_{\Pi} [\mathbb{I}(R; X) - \mathbb{I}(R; X | T_B)] \geq c - d, \quad \text{hence} \quad \liminf_n A_n \geq c - d.$$

The richer architecture recovers the pooled gap up to its own remaining floor. The conclusion is a positive domination statement only when $c > d$. If the richer learner consistently estimates the full conditional residual law, so that $d = 0$, this holds automatically, and for every $\eta > 0$ the advantage eventually exceeds $c - \eta$. Under such a prior, global residual pooling is not Bayes-optimal as a prediction device, even though its conformal intervals retain exact marginal coverage under exchangeability.

This is the relevant sense in which coverage-driven residual prediction may be a dead end. The coverage theorem is true. The architecture nevertheless pays a persistent predictive-information cost that realistic conditional learners can eventually overcome. Conversely, at small samples or under a prior close to $R \perp X$, the pooled architecture can lie near the optimum because its information floor is small and its estimation burden is minimal.

Conformal methods remain compelling when the certificate itself has decision value: selective prediction with abstention, risk control, regulatory reporting, anomaly testing, or safety systems with fallback actions (Angelopoulos et al., 2021; Bates et al., 2021; Angelopoulos et al., 2024b). In those applications the correct comparison is not a proper score alone but the full utility of the certified decision rule. A coverage guarantee does not remove the relevance of predictive scoring, and predictive underperformance does not make a coverage certificate worthless. The decomposition identifies which value is being purchased and what it costs.

12 Conclusion

The exact result is a decomposition. A forecasting architecture exposes a representation T to its predictive law, and under log loss the cost of the restriction is the information gap $\mathbb{I}(Z; X | T)$ plus a within-representation excess. For residual systems the gap of pooling after a state-dependent invertible transformation ϕ_X is $\mathbb{I}(\phi_X(R); X)$, so the irreducible cost of pooled residual prediction is determined by the coordinates in which pooling occurs. Conformal calibration can improve the excess and supply a certificate. It cannot reduce the gap without changing the retained representation.

The empirical result is an attribution. On this benchmark, conditional scale is the principal residual state omitted by raw pooling. The finding survives a matched-predictor comparison, an

i.i.d. simulation of the certificate itself, and replication on an out-of-development holdout. Scale is also the largest component in a same-game Shapley analysis of the surrogate coalition system on 234 series. With the point predictor held fixed, scale-normalized empirical pooling was the best construction we tested under the quantile-approximated CRPS.

The design principle follows. Choose the representation, the residual-law estimator, and the certificate jointly, under the scoring rule and the prior that describe the problem. The useful question is therefore not whether calibration or machine learning is universally superior. It is which representation, estimator, and certificate jointly minimize the decision-maker’s risk under the problems Nature is believed to generate.

Code and data availability

Code and data references are withheld for double-blind review and will be restored on acceptance. The experiments are Python scripts against a public online-forecasting library; script names match the labels used in the text.

A Terminal-leaf bake-off

In the grid of Section 3.2, scale-normalized empirical pooling beat the complete forecaster’s fitted density under the quantile-approximated CRPS, while under the log score the ladder showed the complete forecaster well ahead. This appendix asks the finer, leaf-level question: holding the whitened residual stream fixed, which terminal law should one attach?

Empirical pooling has one limitation that no choice of coordinates removes: it says nothing about tails. The empirical distribution is a statement about where one has data, and it does not extrapolate beyond its observed extremes. High quantiles simply equal the sample maximum. An unsmoothed empirical measure is not directly usable as a continuous log-density forecast, since it assigns zero density off its support. Tail mass is unobserved by definition, so quoting it requires an assumption, whatever machinery carries it. A conformalized parametric or semiparametric model can carry tail mass, with the shape supplied by assumption rather than by distribution-free information.

The conformal literature’s implicit answer is restraint: at calibration levels finer than $1/(n+1)$, distribution-free validity requires an infinite or trivial endpoint, or suitable randomization, so nontrivial finite bounds stop at the calibration resolution. That serves the coverage guarantee and forfeits density evaluation.

The semi-parametric synthesis is to pool the body empirically and model the tail: an empirical law where data is dense, spliced to a peaks-over-threshold tail where it is sparse. Extreme value theory supplies one principled asymptotic family of assumptions for that region. Under CRPS and integrated quantile scores the tail choice matters little, since those scores penalize location errors roughly linearly and have little far-tail density sensitivity. Under the log score it is decisive. This is one more instance of the scoring rule determining the design. We document here a bake-off of terminal leaves, on the same whitened residual stream, that sharpens that finding. The protocol matches the grid: the same 701 series and fixed mean, the same EWMA scale and AR(1) whitening, and CRPS on the common quantile grid. For the log score every leaf is a genuine density. Empirical variants are smoothed by a binned Gaussian kernel with Silverman bandwidth. Generalized Pareto (GPD) tails are fitted by moments to exceedances beyond the 95% thresholds. The conformal tail completion leaves all observed data empirical and shapes

only the reserved $1/(n+1)$ mass with the GPD. Scripts and per-series results accompany the paper.

Terminal leaf	CRPS rel. Gaussian mean / median	log score – Gaussian mean / median (nats/obs)	beats Gaussian (log score)
Gaussian, EWMA variance	1.000 / 1.000	0 / 0	
fitted scale mixture	1.161 / 1.000	+0.10 / +0.011	79%
empirical	0.936 / 0.985	+0.19 / +0.023	84%
windowed empirical (300)	0.936 / 0.986	+0.18 / +0.020	79%
empirical + GPD tails	0.938 / 0.986	+0.21 / +0.034	83%
conformal tail completion	as empirical	+0.21 / +0.033	90%
score-weighted leaf ensemble	0.942 / 0.987	+0.22 / +0.043	98%

In the log-score columns positive numbers are improvements over the Gaussian leaf, in nats per observation. Every leaf’s log score is evaluated as a mixture with a common reference Gaussian (weight 0.01), so all columns are genuine log scores under one convention and no leaf benefits from a private clamp. Means exceed medians because a minority of heavy-tailed series dominate them. All bandwidths, thresholds, mixture weights, and GPD fits are computed from past observations only, so every evaluation is strictly out of sample.

Under the quantile-approximated CRPS every empirical variant beats every fitted law. The differences among empirical variants are a few tenths of a percent, and banded smoothing and an online quantile tracker are indistinguishable from the plain empirical. Under a score with little far-tail sensitivity the estimator class is what matters, and refinements are cosmetic.

Under the log score the empirical leaves win as well, and the tails decide the margins. The fitted mixture trails every empirical variant, the empirical body with GPD tails leads on mean gain, and the conformal tail completion has the best win rate among single leaves (90%) and is strongest on short series (median gain +0.039 nats per observation on series shorter than 1200, improving 86% of them, against +0.015 on series longer than 2500). The best overall leaf is a decayed-score-weighted ensemble of the leaves themselves, which beats the Gaussian on 98% of series. A windowed empirical adds nothing over the expanding one at any length: forgetting belongs in the scale state, not the leaf. Which finite-sample validity properties survive when reserved mass is shaped by a fitted tail model is not established here, and the tail-completion construction should be read as a predictive proposal, not a certified one.

The bake-off compares estimator classes on a fixed whitened stream, and it invites a package-level conclusion that turns out to be wrong. We implemented the empirical leaf and the score-weighted leaf ensemble as terminal options inside the complete forecaster and reran the comparison in situ, with every other component at its production setting (`leaf_insitu.py`). The swap improved nothing. On the same 701 series the empirical leaf lost to the reference configuration under both scores (relative CRPS 1.03, winning 14% of series; log score -0.26 nats per observation, winning 6%), and the ensemble leaf was closer but still behind. The reconciliation is the three-price ledger of Section 3: the reference system already realizes the empirical advantage through its CRPS-fitted terminal leaf and its tail splice, so the model-form gain the bake-off measures has already been spent. The idea of residual pooling did not, in this case, improve the shipped model even after all the conditional modeling was done.

B Deferred proofs

Proof of Proposition 1. Apply Lemma 3 conditionally on D_n at rung j : the log risk of $\hat{Q}_j$ splits as

$$\hat{\mathcal{L}}(\hat{Q}_j) = H(Y | T_j, D_n) + \epsilon_j(D_n),$$

oracle entropy plus within-representation excess, and likewise at rung $j - 1$. Subtracting,

$$\Delta_j = [H(Y | T_{j-1}, D_n) - H(Y | T_j, D_n)] + \epsilon_{j-1}(D_n) - \epsilon_j(D_n).$$

Because T_{j-1} is a function of T_j , conditioning on T_j is the same as conditioning on (T_j, T_{j-1}) , so

$$H(Y | T_{j-1}, D_n) - H(Y | T_j, D_n) = H(Y | T_{j-1}, D_n) - H(Y | T_j, T_{j-1}, D_n) = I(Y; T_j | T_{j-1}, D_n),$$

which is Corollary 6 in entropy form.

Proof of Proposition 2. Sum (3) over $j = 1, \dots, m$. The excess-risk terms telescope to $\epsilon_0(D_n) - \epsilon_m(D_n)$. For the information terms, write each as an entropy difference as in the previous proof and telescope again:

$$\sum_{j=1}^m I(Y; T_j | T_{j-1}, D_n) = \sum_{j=1}^m [H(Y | T_{j-1}, D_n) - H(Y | T_j, D_n)] = H(Y | T_0, D_n) - H(Y | T_m, D_n).$$

By Corollary 6 applied to the pair (T_0, T_m) with $T_0 = g(T_m)$,

$$H(Y | T_0, D_n) - H(Y | T_m, D_n) = I(Y; T_m | T_0, D_n) = I(Y; X | T_0, D_n) - I(Y; X | T_m, D_n),$$

which gives (4). Adding and subtracting $I(Y; X | T_m, D_n)$ in the telescoped sum gives the remainder form (5).

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